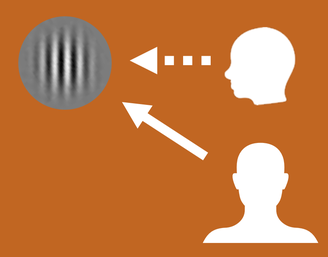
Spatial cuing mechanisms behind visual perspective-taking



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INTRODUCTION

- We commonly take other's perceptual perspectives: we represent what others can see
- Someone's congruent (vs incongruent) perspective can facilitate (vs impair) one's **perceptual performance** to detect stimuli in a scene, inducing an "altercentric" **congruence effect**[1,2]
- We assume that this influence occurs by shifting spatial attention to the target of the other's perspective (even when this shift is mediated by mentalising!)

What are the spatial cuing mechanisms behind visual perspective-taking effects on perceptual performance?

HYPOTHESES

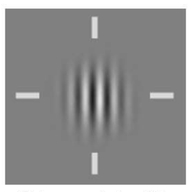
Two possible (not mutually exclusive) mechanisms behind altercentric influence. The avatar's cue…

where?

what?

- **(H1) Reduces spatial uncertainty** before signal processing: induces prior expectations about target location
- **(H2) Increases signal-to-noise processing** at the cued location: over and beyond facilitating localisation, it boosts signal processing at one location

- We manipulate spatial uncertainty with **visible markers** on stimulus location.
- Non-social cues studies: when stimuli are well-localised with markers, the congruence effect perceptual performance **disappears**[3]



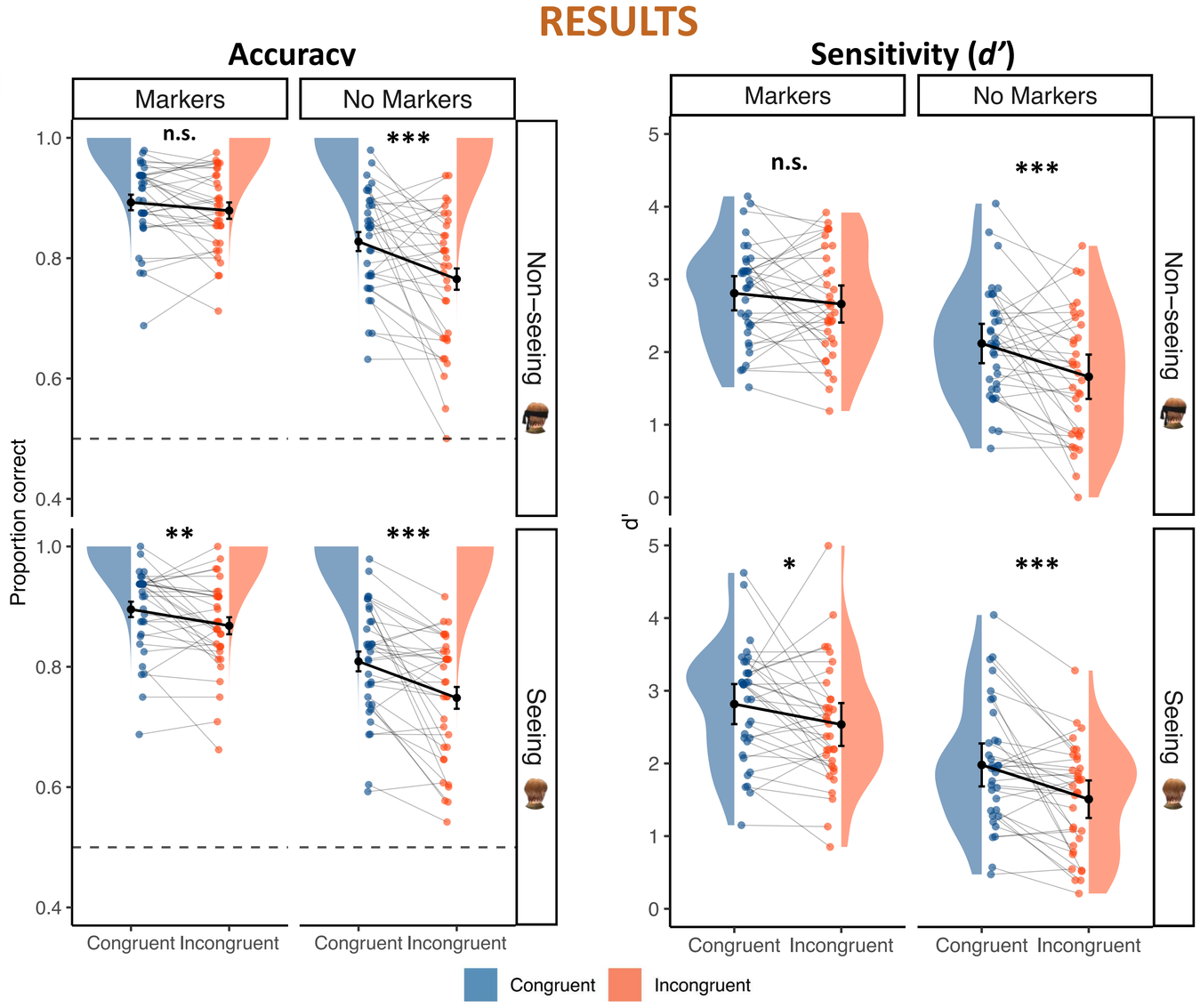
Gabor patch with localising markers

- **(VPT) Visual-perspective specific hypothesis:** Congruence effects are due to visual perspective-taking: *sensitivity to the avatar's visual state/mentalising*, rather than low-level directional cues.

Predictions

(H1) Congruence effects are absent or lower for Marker (vs No-Marker), as stimuli as easily located.
(H2) Congruence effects survive Marker/No-Marker manipulation, impacting processing beyond localisation.
(VPT) As VPT mechanisms differ or augment those of directional-cuing, we expect to find support for H2 only in sighted avatars. (Interaction: *Congruence* X *Markers* X *Sight*)

RESULTS



METHODS

Online **pilot** run with jsPsych. N=48 recruited on Prolific; **34** after exclusion (16 Female, M=33.32y., SD=11.89).

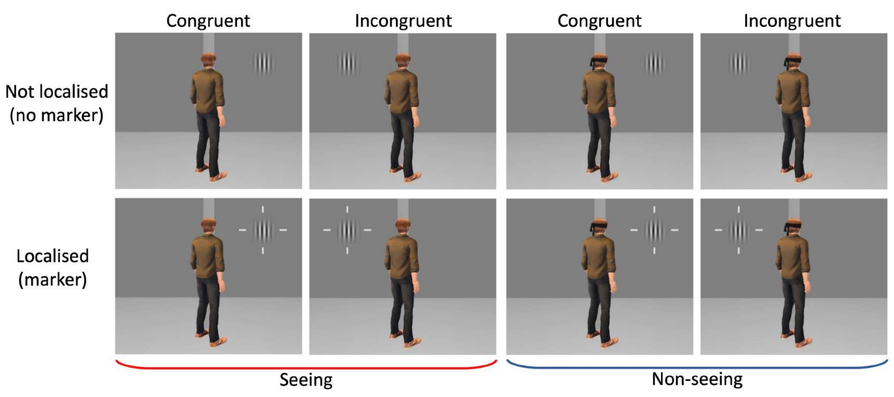
Task: **Orientation task**: Is the target oriented **vertical** or **horizonal**?[3]

Stimuli: **Noise**: 10% contrast white noise
Target: Gabor over noise oriented vertically/horizontally, contrast staircase-adjusted for 75% accuracy

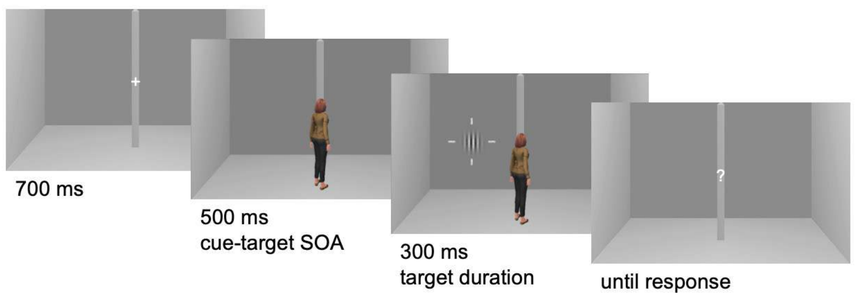
Familiarisation: Prior to main task, subjects completed 24 trials responding whether the avatar can "see" the Gabor or not

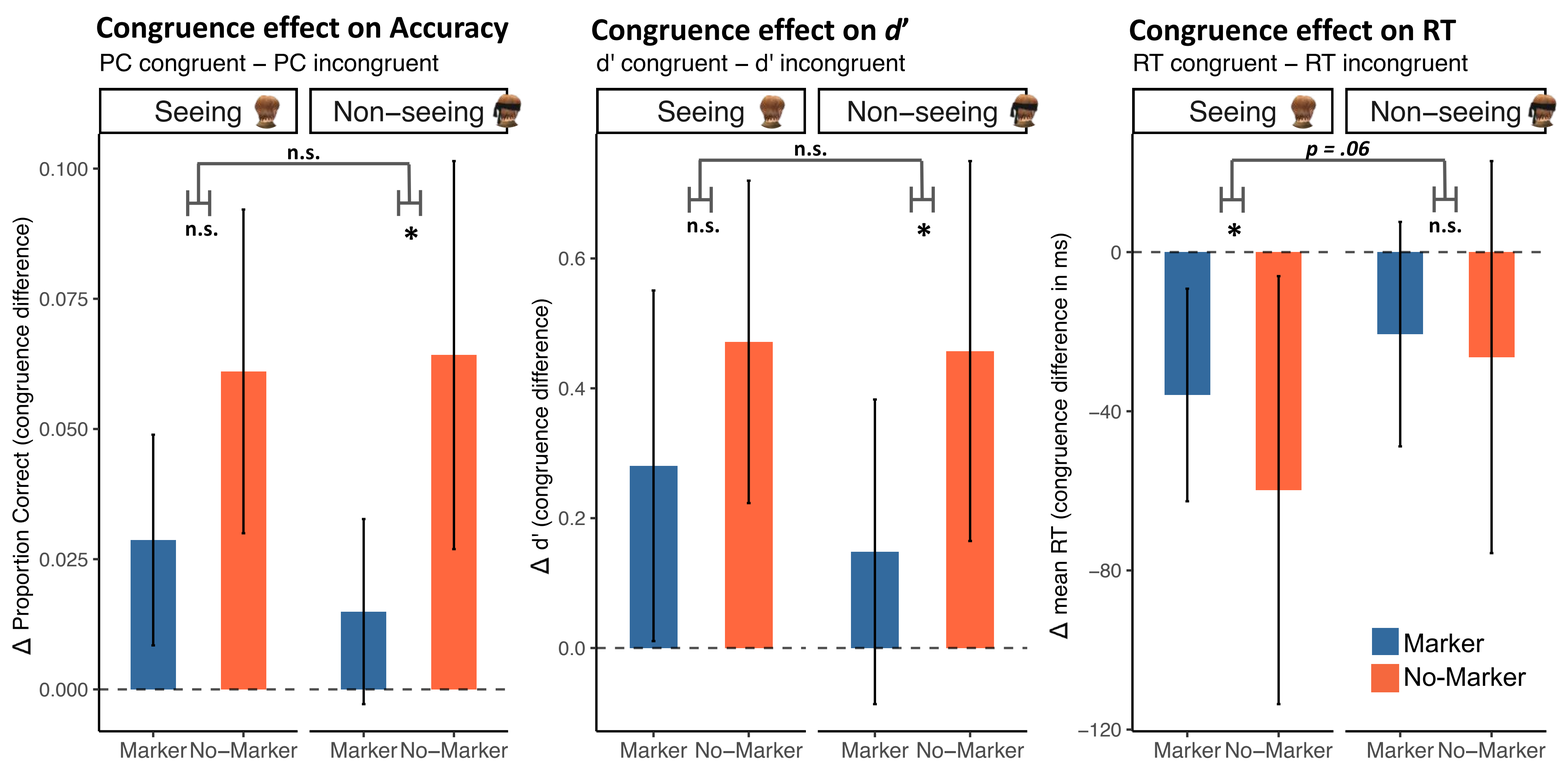
Conditions: Within-subjects 2 x 2 x 2:

- **Localising Marker Presence** (Marker vs No-Marker)
- **Avatar Orientation Congruence** (Congruent vs Incongruent)
- **Avatar Sight Condition** (Seeing vs Non-seeing)



Procedure: 48 trials per condition: 384 total trials randomized





CONCLUSIONS

- **Congruence** and **Marker** have a significant main factor boosting effects on accuracy and sensitivity (*d'*)
- Support for **H1 in Non-seeing avatar:** when stimuli are easily identified, congruence effects are lower, suggesting that the avatar's cue mainly to **reduce spatial uncertainty**, helping localise the target
- Non-significant **small tendency** towards supporting **VPT**: Congruence effects in accuracy and *d'* survive Marker manipulation **only in Seeing avatars**, suggesting a boost in processing at cued location beyond spatial uncertainty. But the overall effect was not significant (Congruence X Markers X Sight: all *ps* > .05).
- Next steps: Follow-up this pilot with larger N study!
  Investigate whether different cue-target SOAs produce similar results

[1] Seow, T., & Fleming, S. M. (2019). Perceptual sensitivity is modulated by what others can see. *Attention, Perception, & Psychophysics, 81*(6), 1979–1990.
[2] Battich, L., Pacherie, E., & Grèzes, J. (2025). Social perspective-taking influences on metacognition. *Cognition*, *254*, 105966.
[3] Gould, I. C., Wolfgang, B. J., & Smith, P. L. (2007). Spatial uncertainty explains exogenous and endogenous attentional cuing effects in visual signal detection. *Journal of Vision*, *7*(13), 4.

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